Janushan Sivakrishna

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Education

Toronto Metropolitan University Sept. 2023 – Apr. 2028
Bachelor of Engineering – Mechanical Engineering (B.Eng.) *Toronto, ON*

- **Leadership:** Ontario Engineering Competition (Competition Lead), Mechanical Contractors Association of Toronto TMU Chapter (President), MECU (VP Finance)
- **Relevant Courses:** Materials Science, Solid Mechanics, Applied Thermodynamics, Control Systems

Technical Skills

CAD & Design: SolidWorks (CSWA), QuickSurface, AutoCAD, Fusion 360, Onshape, Parametric Modeling, Surface Modeling, Assemblies, Engineering Drawings, Bill of Materials (BOM)
Reverse Engineering & Manufacturing: 3-D Scan-to-CAD, Mesh Repair, STL Export, 3-D Printing (FDM), Design for Additive Manufacturing, CNC Machining, Rapid Prototyping
Analysis & Tools: Ansys Mechanical (FEA), MATLAB, Python, C/C++, Microsoft Excel (VBA), SAP, Raspberry Pi, Arduino

Experience

Product Design Engineer June 2026 – Present
CURRENT Engineering Studio *Brampton, ON*

- Reverse-engineered mechanical components using QuickSurface and SolidWorks, converting 3-D scan data into accurate parametric CAD models for manufacturing and redesign.
- Processed and repaired scan mesh data, achieving dimensional accuracy within ±0.15 mm between scanned geometry and reconstructed CAD models.
- Created engineering drawings and assemblies from reverse-engineered components, reducing redesign time and supporting rapid prototyping.

Mechanical Design Intern Mar. 2025 – May 2025
CPM Tools Industries Inc. *Toronto, ON*

- Optimized three self-adjusting and ratchet mechanisms in SolidWorks, improving tool efficiency by 10%.
- Produced detailed technical drawings and a comprehensive BOM in SolidWorks, ensuring full manufacturing accuracy and material traceability.
- Conducted competitive analysis of four competitors and wrote a 5-page market expansion report identifying three new product opportunities.

Senior Technical Student Sept. 2026 – Present
Toronto Hydro *Toronto, ON*

- Helped design energization, upgrades, and removal of electrical services for hundreds of high-capital construction projects across the Toronto power grid, supporting high-level design and project management.
- Developed project and finance tracking methods for the department, supporting six managers; built an Excel VBA macro with SAP connection and billing modules to automate billing in SAP Enterprise (ML81N).

Projects

Honda Accord Ducktail Spoiler Oct. 2026 - Present

- Designed a custom ducktail spoiler in SolidWorks, using QuickSurface to convert a 3-D scan of the trunk lid into CAD surfaces that follow the car's contours; split the 1.17 m span into 5 segments to fit a 256 mm Bambu Lab A1 print bed.
- Engineered a flush-mount third brake light recess with a hidden wire channel to the trunk, driven by equation-based global variables for parametric resizing to different light modules; applied 0.2 mm print-fit clearances.
- Designed doweled joints (12 custom Ø8 mm printed pins) and hidden engraved part numbers for assembly; exported and verified watertight STL meshes for FDM batch printing.

Camera-Guided Following RC Car Sept. 2026 – Present

- Designed a SolidWorks chassis and component mounts, 3-D printed all parts, and built a camera-guided RC car (Raspberry Pi Zero 2 W, OV5647) that tracks an ArUco marker with OpenCV and estimates distance to within ±0.5 cm from 10 to 140 cm at 25 FPS.
- Programmed Python distance-keeping logic (30 cm gap, 25 cm hard stop) for an L298N four-motor differential drive, with an alpha-beta filter, a 0.3 s camera-fault watchdog, and a Flask web dashboard with live telemetry and tuning.

Rankine Cycle Plant Design Oct. 2025 – Dec. 2025

- Designed an 80 MWe / 25 MWth natural gas Rankine cycle power plant to provide sustainable power and district heating for remote Canadian communities. Focused on maximizing thermal efficiency and economic viability through advanced cycle optimization.
- Engineered a four-stage turbine setup with reheat and regenerative cycles; compared three iterations.
- Validated with MATLAB simulations using the XSteam library; performed methane cost analyses.